

Ria S.

Software Developer · Data Science

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PROFILE SUMMARY

Software Engineer and Data Science student with demonstrated experience shipping full-stack, cloud-deployed applications. Adept at integrating ML/AI pipelines, computer vision, and AWS infrastructure into production systems, with a strong interest in low-level systems. Passionate about building intelligent, scalable solutions — from wardrobe AI to an x86 operating system written from scratch.

EDUCATION

B.Tech — Computer Science & Data Science

2024 – 2028

Manipal University Jaipur · Jaipur

WORK EXPERIENCE

Data Science & Analytics Intern

Oct 2025 – Dec 2025

Future Interns — Remote

- ▶ Conducted end-to-end analysis on e-commerce and social media datasets; delivered insights on customer behaviour trends.

Software Engineering Job Simulation

Jul 2025

J.P. Morgan · Forage

- ▶ Built financial data tooling prototypes; worked with real-world engineering workflows using JPMorgan's Perspective library.

Data Analytics Virtual Experience

Jun 2025

Quantium · Forage

- ▶ Analysed retail transaction data to uncover purchasing trends and communicate strategic recommendations.

Data Analytics Virtual Internship

Jun 2025

Deloitte · Forage

- ▶ Prepared, modelled, and visualised forensic data for business storytelling using Tableau and Excel.

PROJECTS

WYA — What's Your Aesthetic

Jan 2025 – May 2026

- ▶ Built an end-to-end AI pipeline: background removal (rembg + OpenCV) → garment masking (GrabCut) → zero-shot classification via AWS SageMaker (FashionCLIP) → KMeans colour extraction → pattern detection → rule-based fabric classifier.
- ▶ Delivered 10+ features including Style DNA Quiz, AI Outfit Matcher, Green Score, Vacation Packer, and weather-based styling.
- ▶ Architected the AWS deployment: React + TypeScript on S3 + CloudFront; FastAPI + Docker on EC2; SageMaker serverless endpoint; CI/CD via GitHub Actions; rate-limited AI endpoints; 47 pytest tests.
- ▶ Built **Luna**, a conversational stylist on top of the WYA API: a regex intent classifier routes structured queries to WYA endpoints, with an OpenRouter LLM fallback that receives the user's real wardrobe as context. Separate FastAPI/Docker/EC2 deployment with JWT auth shared with WYA.

Stack: Python · FastAPI · React · TypeScript · AWS (EC2, S3, CloudFront, SageMaker) · Docker · OpenCV · SQLite

Lumen — 32-bit x86 Operating System

Present

- ▶ Built an OS from a raw boot sector up: custom BIOS bootloader and a freestanding ~13,000-line C kernel.
- ▶ Implemented paging with per-task address spaces, a bitmap frame allocator, an interrupt-safe heap, and preemptive Ring 0/Ring 3 multitasking with fork/exec.
- ▶ Added a 25-call syscall table, an ATA driver, a persistent filesystem (LumenFS v2), an ELF loader, and a text-mode GUI.
- ▶ Wrote a **make test** harness that boots the OS headless in QEMU and fails on any self-test failure, wired into GitHub Actions.

Stack: C · x86 Assembly · NASM · QEMU · GitHub Actions

TryOn — Build, Compare, and Perfect Outfits

May – Jun 2026

- ▶ Built a computer vision pipeline that turns any clothing screenshot into a transparent garment cutout (rembg), auto-categorizes it with a FashionCLIP zero-shot classifier, and fits it onto a 2D paper-doll or rigged 3D mannequin using MediaPipe BlazePose landmarks and thin-plate-spline warping.
- ▶ Built the 3D Mannequin Garment Viewer: a CV analyzer detects strap type and back construction from one photo, with a 75% confidence gate that reports “undetermined” rather than hallucinating a back. Falls back to a client-side Canvas engine when the backend is offline.
- ▶ Tested the FastAPI backend with 69 tests at 86% coverage, run against the real CV models instead of mocks.

Stack: Python · FastAPI · React · TypeScript · React Three Fiber · MediaPipe · rembg · SQLite

FairFaceGuard — Deepfake Detection Under Skin-Tone and Lighting Variations

Present

- ▶ Designed a causal research pipeline testing whether a deepfake detector's accuracy gap across skin-tone groups comes from skin tone itself or from correlated illumination cues.
- ▶ Built illuminant-corrected ITA/Fitzpatrick labelling and matched counterfactuals (skin-only, light-only, both) with an automated manipulation check.
- ▶ Interrogated a frozen EfficientNet-B4 baseline via adversarial GRL sweeps, linear probing, and a counterfactual causal test, with bootstrap CIs and a paired permutation test. Pilot run on a FaceForensics++ subset.

Stack: Python · PyTorch · EfficientNet-B4 · FaceForensics++

SKILLS & TECHNOLOGIES

Languages: Python · TypeScript · JavaScript · Java · C · x86 Assembly · SQL (MySQL)

Frameworks & Libraries: FastAPI · Express.js · React · Node.js · OpenCV · MediaPipe · scikit-learn · PyTorch · Pandas · NumPy

Cloud & DevOps: AWS (EC2, S3, CloudFront, SageMaker, IAM) · Docker · GitHub Actions · CI/CD

Data & ML: Data Analysis · KMeans Clustering · Computer Vision · GNNs · Zero-shot Classification · Regression

Databases: SQLite · MySQL

Tools: Git · Pytest · QEMU · Vite · Tableau · Excel

CERTIFICATIONS

- ▶ IBM Data Analyst Professional Certificate
- ▶ Google Data Analytics Professional Certificate
- ▶ Google IT Automation with Python
- ▶ Meta Data Analyst with GenAI
- ▶ Programming in Java — NPTEL

LANGUAGES

English (Native) · Hindi (Advanced) · Malayalam (Beginner)